

BLDC MOTOR With Web-Based Speed Control Using ESP32

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Integrating wireless control into brushless direct current (BLDC) motor systems opens up exciting possibilities for applications such as remote-controlled cars, robots, and other innovative systems. BLDC motors are commonly operated using a driver or a remote-controlled controller paired with an electronic speed controller (ESC). Adding Wi-Fi connectivity to enable control via a laptop or personal computer further enhances their versatility.

This motor system utilises a BLDC motor, an ESC, and an ESP32 to enable wireless operation. Bill of Materials table lists the required components, while the author's prototype of the BLDC motor controller is depicted in Fig. 1.

An ESC is vital in operating BLDC motors. It regulates the motor's speed, direction, and overall performance by controlling the power delivered to the windings.

The 3S 12V battery powers the BLDC motor and other components, ensuring efficient system operation. Its stable output makes it suitable for driving ESCs and maintaining

consistent power for the ESP32 module. Fig. 2 shows an electronic speed controller.

Working principle

The ESC manages the timing and duration of current flow through the stator windings, effectively controlling the rotor's rotation. It operates on principles such as pulse width modulation (PWM), phase sequence, sensorless control, and sensor-based control.

The ESC uses PWM signals to control transistors, managing

BILL OF MATERIALS		
Components	Quantity	Description
ESC controller	1	30A Simonk ESC controller
NodeMCU ESP32 (MOD1)	1	ESP32 development kit
Battery	1	3S 12V battery
Jumper	2	Female to male jumper

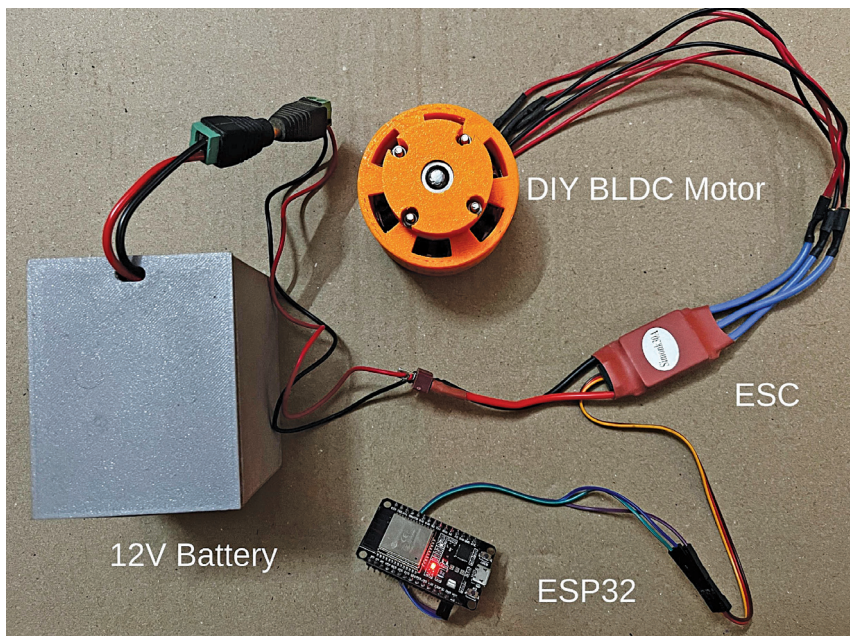


Fig. 1: Authors' prototype



Fig. 2: An electronic speed controller (ESC)



Fig. 3: ESP32 (NodeMCU)